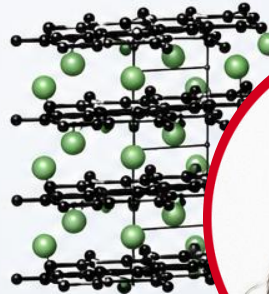
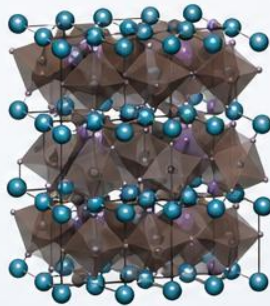
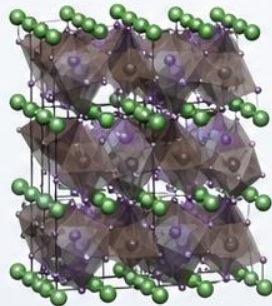
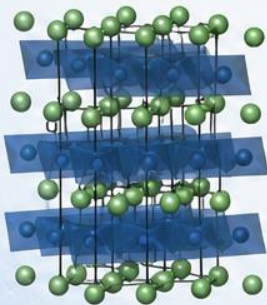


Quantum-Chemical Bonding Analysis of Functional Solid-State Materials (with Emphasis on Batteries)



Dr. Christina Ertural

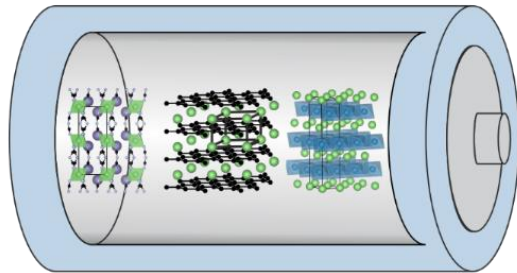
Theoretical Chemistry & Materials Informatics

Outline

- Motivation & short introduction to LOBSTER
- Covalency & Ionicity
 - how it can be used to assess the stability of materials and find new suitable candidates
- Alternative battery materials
 - amorphous materials for anodes
 - transition metal free cathodes

Academic background

- Study of chemistry (B. Sc. & M. Sc. & PhD) at RWTH Aachen University
- Field of research in Prof. Dronskowski's group at Inorganic Institute of RWTH Aachen:
 - *Theoretical and quantum chemistry*
 - > Density functional theory
 - *Chemical bonding* in functional materials
 - > Solid-state chemistry and materials
 - Focus: Method & *C++ code development* concerning chemical bond descriptors in LOBSTER
 - > Mulliken & Löwdin charges, Madelung energy, band-resolved COHP, COBI



Academic background

- Field of research in Prof. George's group at Bundesanstalt für Materialforschung und -prüfung (BAM) Berlin:
 - *Materials informatics*
 - > automation, workflows & machine learning
 - *AutoPLEX*
 - (automatic potential-landscape explorer)
 - > software for generating and benchmarking *machine learning* (ML)-based interatomic potentials
 - Focus: *Python code development* of AutoPLEX from scratch
 - > Materials Project framework

```
from mp_api.client import MPRester
from autoplex.auto.phonons.flows import CompleteDFTvsMLBenchmarkWorkflow

mpr = MPRester(api_key='YOUR_MP_API_KEY')
structure_list = []
benchmark_structure_list = []
mpids = ["mp-22985"]
# you can put as many mpids as needed e.g. mpids = ["mp-22985", "mp-1185319"] for all LiCl ent
mpbenchmark = ["mp-22985"]
for mpid in mpids:
    structure = mpr.get_structure_by_material_id(mpid)
    structure_list.append(structure)
for mpbm in mpbenchmark:
    bm_structure = mpr.get_structure_by_material_id(mpbm)
    benchmark_structure_list.append(bm_structure)

complete_flow = CompleteDFTvsMLBenchmarkWorkflow(
    apply_data_preprocessing=True,
).make(
    structure_list=structure_list, mp_ids=mpids,
    benchmark_structures=benchmark_structure_list, benchmark_mp_ids=mpbenchmark)

complete_flow.name = "tutorial"
autoplex_flow = complete_flow
```



From theory to chemistry...

Theoretical framework

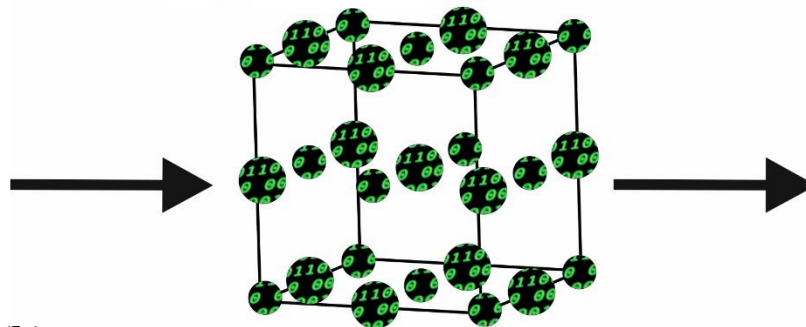
Schrödinger equation

$$\hat{H}\psi = E\psi$$

$$|\psi_j(\mathbf{k})\rangle = \sum_{\mu} c_{\mu j}(\mathbf{k}) |\chi_{\mu}(\mathbf{k})\rangle$$

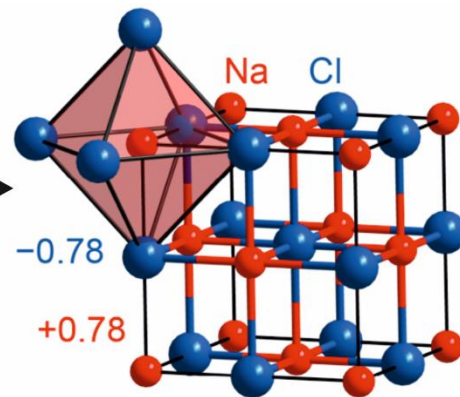
$$\text{COHP}_{\mu\nu} = H_{\mu\nu} \sum_{j,\mathbf{k}} c_{\mu j}^*(\mathbf{k}) c_{\nu j}(\mathbf{k})$$

$$q_A = N - \sum_{\mu \in A} \text{GP}_{\mu}$$



Computer simulation
based on experimental
structure models

Chemical insights
Structure, charges,
covalency, stability...



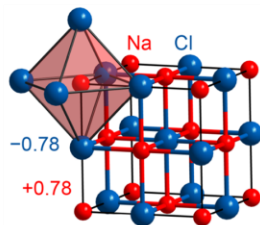
...to (everyday) materials

MULLIKEN and LOEWLIN POPULATION ANALYSIS

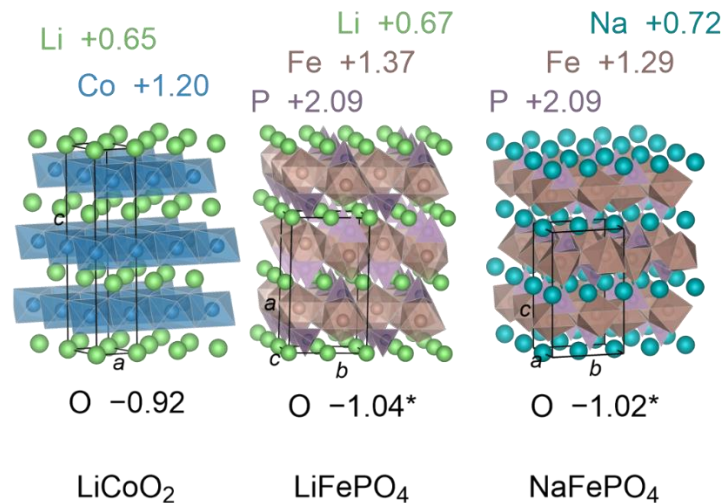
atom#	atom	Mulliken charge	Loewlin charge
1	Na	0.78	0.67
5	Cl	-0.78	-0.67
	total	0.00	0.00

Mulliken and Loewlin gross orbital population

atom#	atom	basisfunction	Mulliken GP	Loewlin GP
1	Na	3s	0.22	0.33
		2p_y	2.00	2.00
		2p_z	2.00	2.00
		2p_x	2.00	2.00
		total	6.22	6.33
5	Cl	3s	1.95	1.87
		3p_y	1.95	1.93
		3p_z	1.95	1.93
		3p_x	1.95	1.93
		total	7.78	7.67

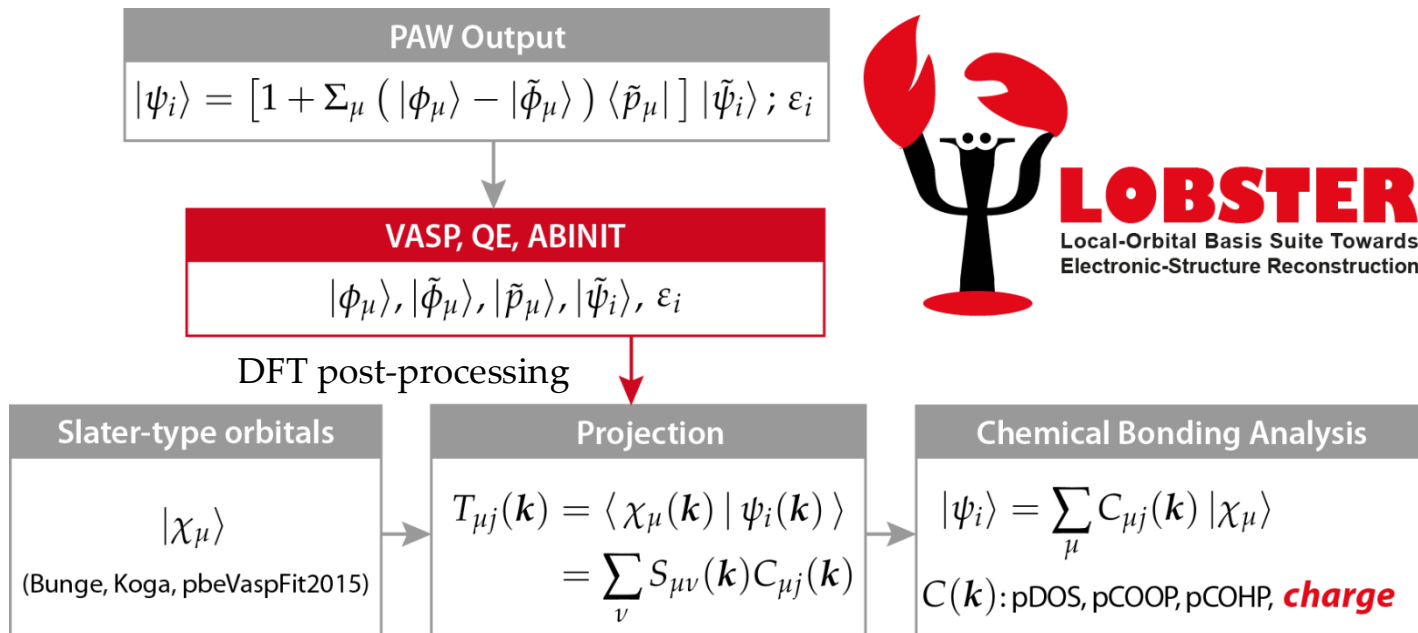


Chemical insights
Structure, charges,
covalency, stability...

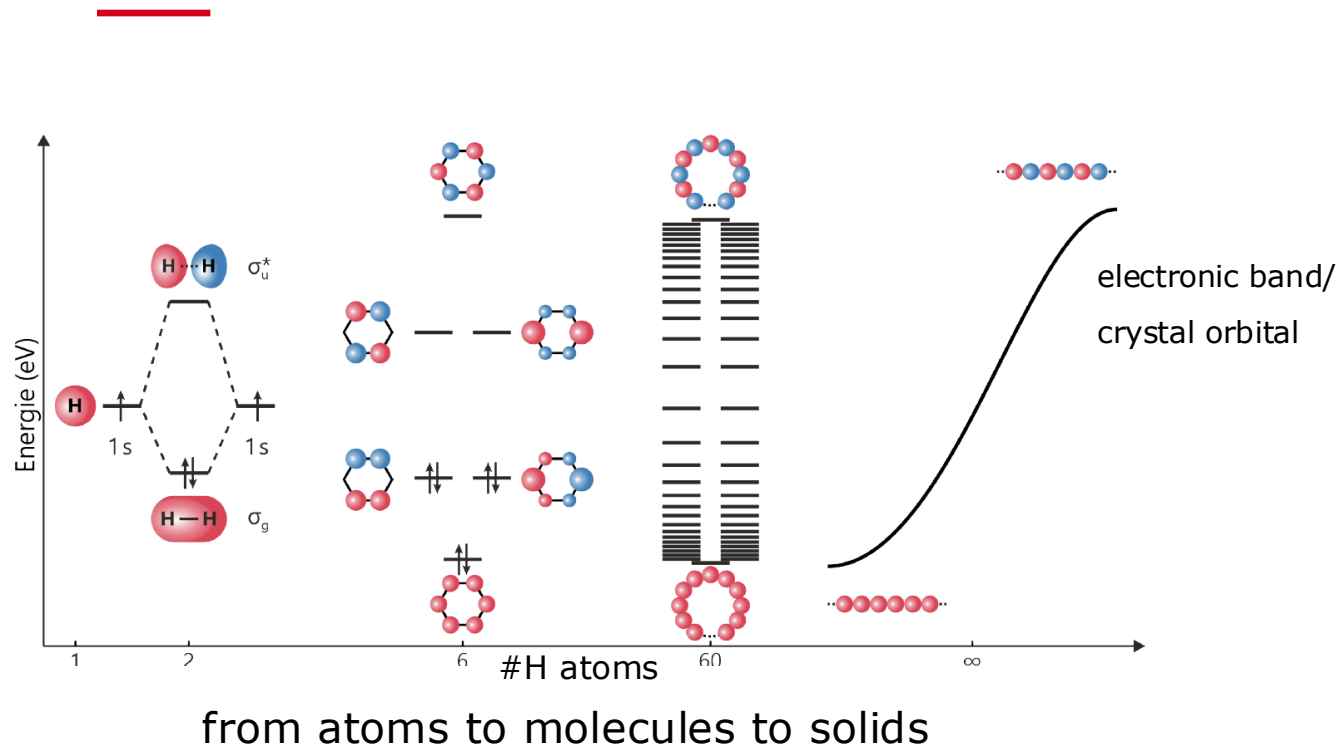


Materials properties
Conductivity, hardness,
(dis)charge behaviour...

The right tool: LOBSTER

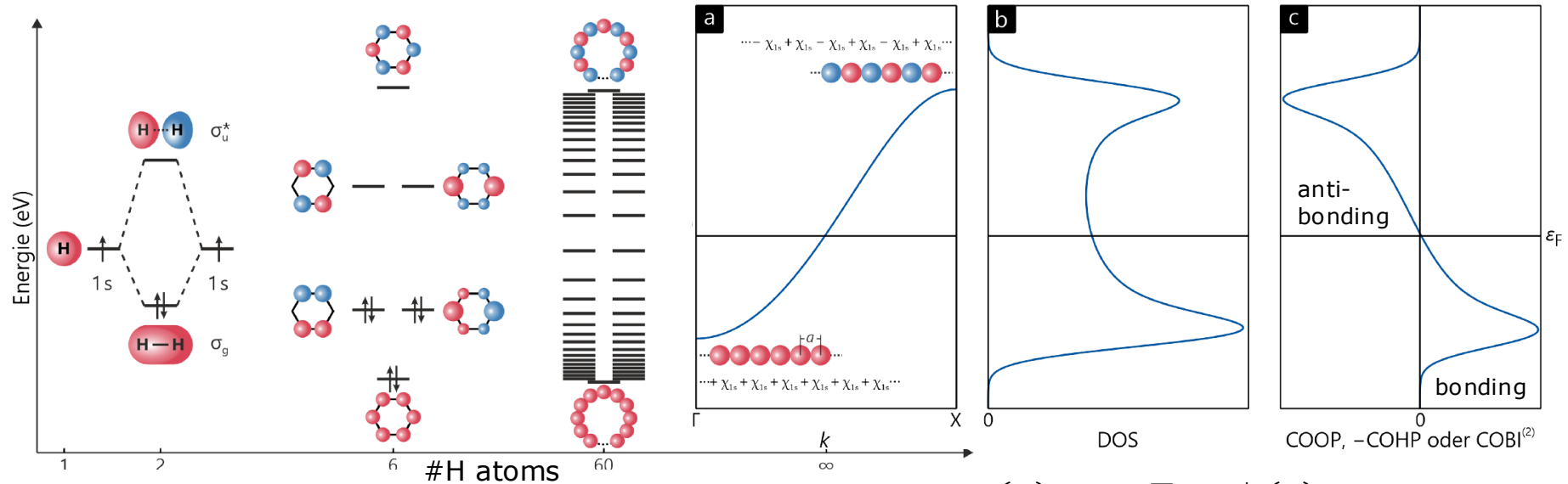


Covalent bond analysis: COOP, COHP



Covalent bond analysis: COOP, COHP

Crystal Orbital Overlap Population & Crystal Orbital Hamilton Population



from atoms to molecules to solids

$$COOP_{\mu\nu}(E) \sim S_{\mu\nu} \sum_{j,\mathbf{k}} C_{\mu j}^*(\mathbf{k}) C_{\nu j}(\mathbf{k})$$

$$COHP_{\mu\nu}(E) \sim H_{\mu\nu} \sum_{j,\mathbf{k}} C_{\mu j}^*(\mathbf{k}) C_{\nu j}(\mathbf{k})$$

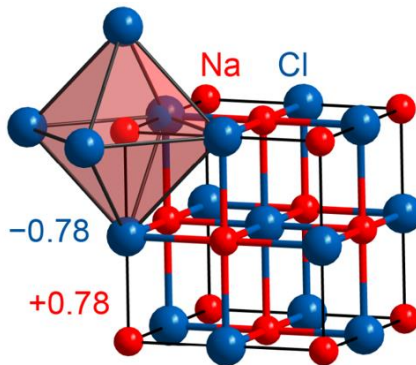
Ionic bond analysis

Mulliken & Löwdin population analysis

- wave function based
- basis set dependent (molecular case)

Popular alternative: Bader charge analysis

- electron density based



LCAO ansatz ($\mathbf{k}$ -space):

$$|\psi_j(\mathbf{k})\rangle = \sum_{\mu} c_{\mu j}(\mathbf{k}) |\chi_{\mu}(\mathbf{k})\rangle$$

Gross (orbital) population:

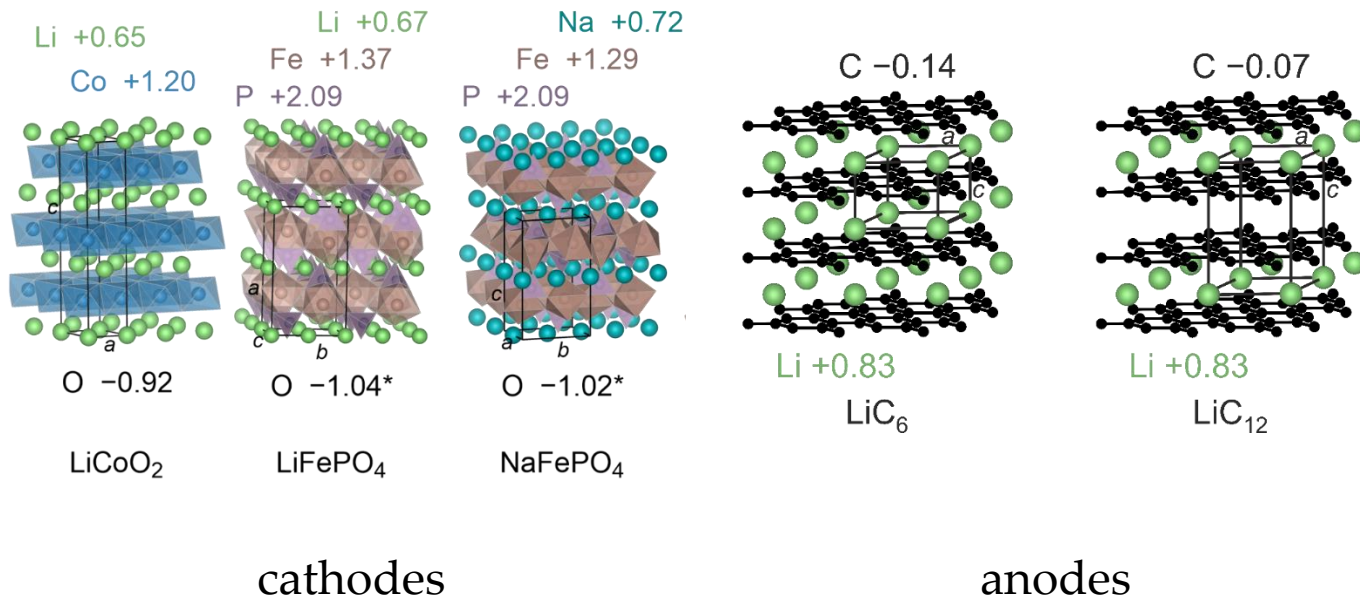
$$GP_{\mu} = \sum_{\mathbf{k}} \sum_{\nu} P_{\mu\nu}(\mathbf{k}) S_{\mu\nu}(\mathbf{k}) w(\mathbf{k})$$

Mulliken/Löwdin charge:

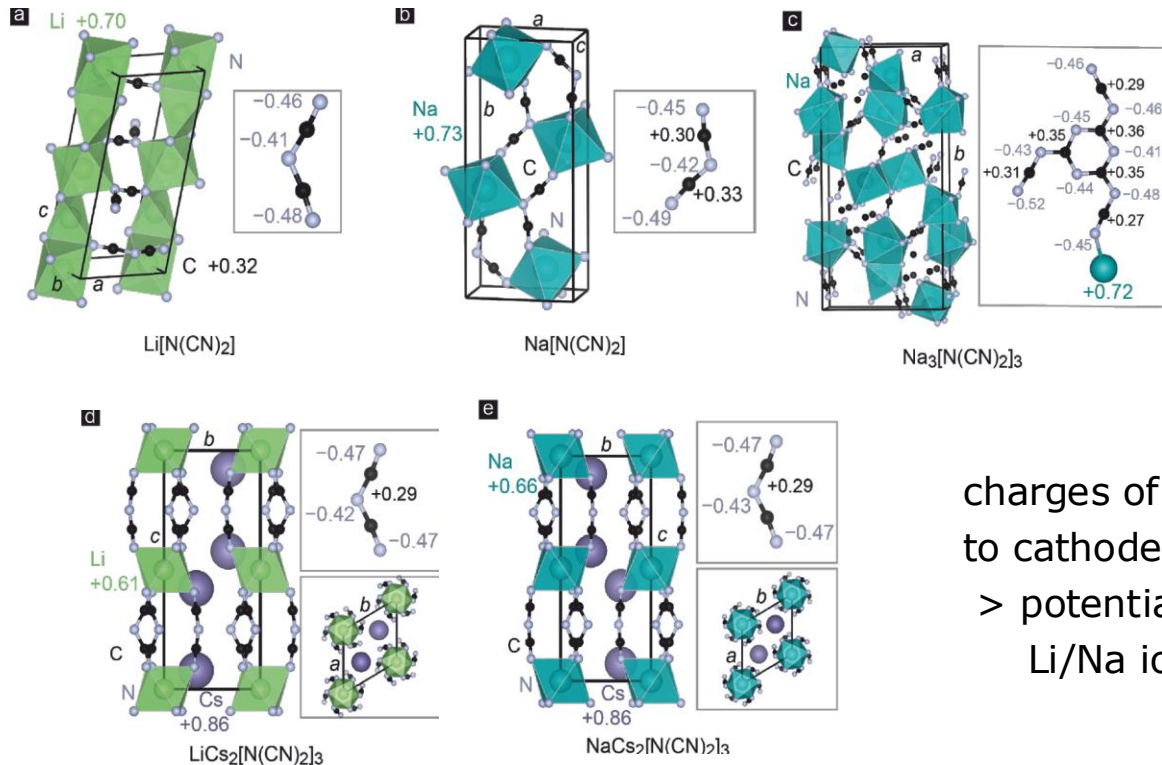
$$q_A = N - \sum_{\mu \in A} GP_{\mu}$$

Mulliken & Löwdin charges

- commercially used materials



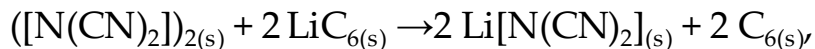
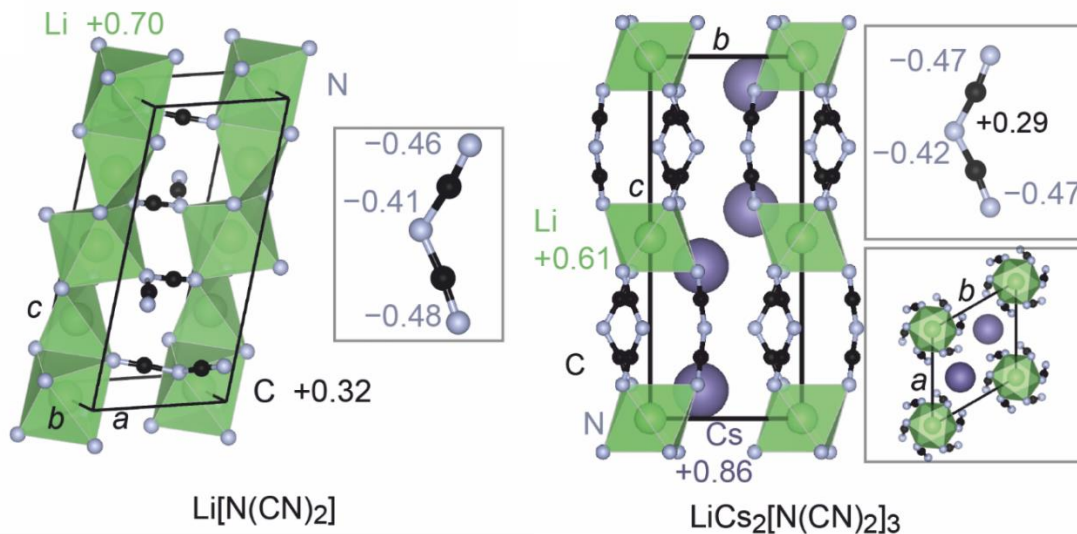
Mulliken & Löwdin charges: Li/Na dicyanamide salts



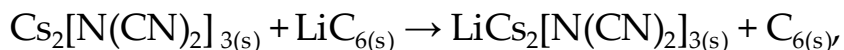
charges of Li and Na comparable
to cathode materials
> potential candidates for
Li/Na ion battery cathodes

Mulliken & Löwdin charges

- new transition metal free candidates



$$\Delta H_r = -390.8 \text{ kJ mol}^{-1} \triangleq \textbf{+4.1 V}$$

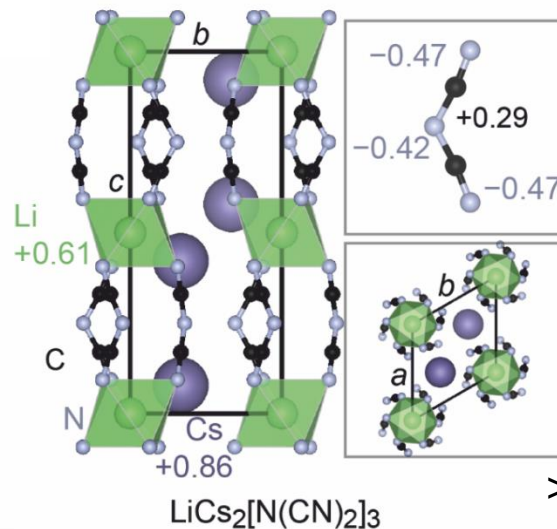
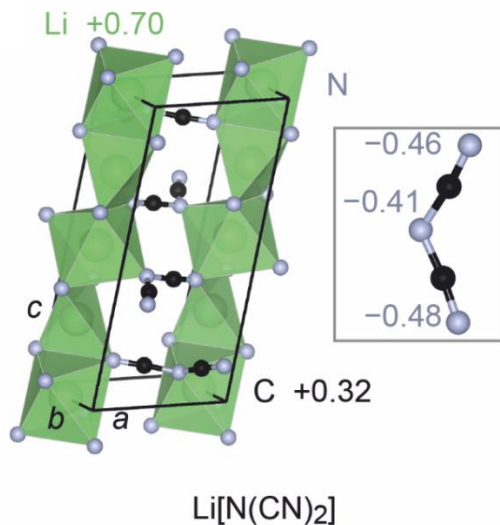


$$\Delta H_r = -444.4 \text{ kJ mol}^{-1} \triangleq \textbf{+4.6 V}$$

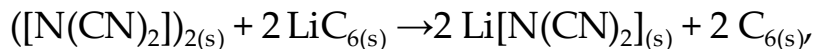
Ertural, C.; Stoffel, R. P.; Müller, P. C.; Vogt, C. A.; Dronskowski, R., First-Principles Plane-Wave-Based Exploration of Cathode and Anode Materials for Li and Na-ion Batteries involving Complex Nitrogen-based Anions. *Chem. Mater.* **2022**, *34*, 652–668.

Mulliken & Löwdin charges

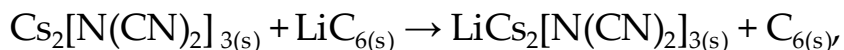
- new transition metal free candidates



> experimental validation ongoing



$$\Delta H_r = -390.8 \text{ kJ mol}^{-1} \triangleq \textbf{+4.1 V}$$

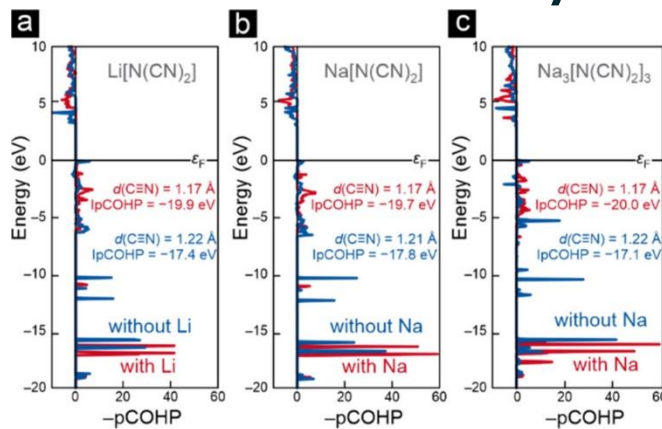


$$\Delta H_r = -444.4 \text{ kJ mol}^{-1} \triangleq \textbf{+4.6 V}$$

Ertural, C.; Stoffel, R. P.; Müller, P. C.; Vogt, C. A.; Dronskowski, R., First-Principles Plane-Wave-Based Exploration of Cathode and Anode Materials for Li and Na-ion Batteries involving Complex Nitrogen-based Anions. *Chem. Mater.* **2022**, 34, 652–668.

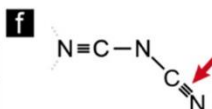
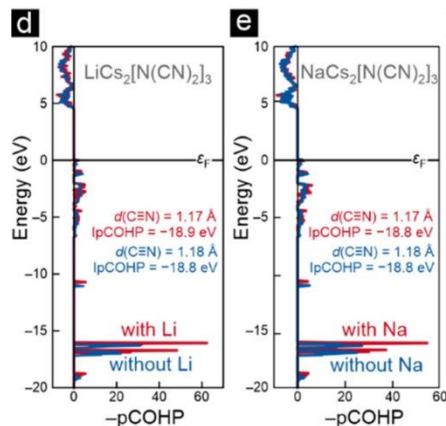
COHP: Deintercalation of Li/Na dicyanamide salts

a lot of
change >



COHP (as measure of chemical bond covalency) plot gives hint on structural deformation and change of chemical environment during deintercalation

almost no
change >



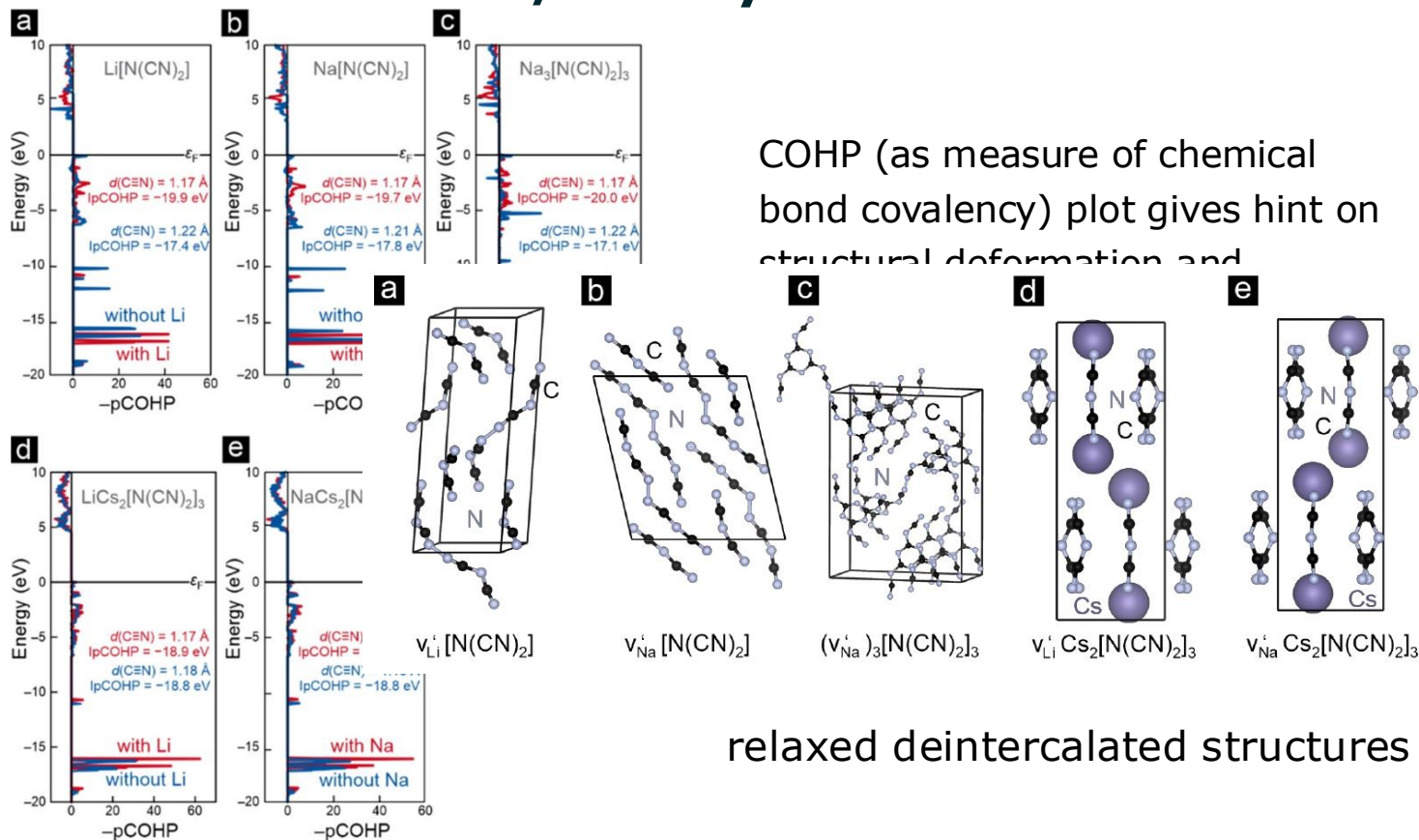
> using COHP to assess stability

Ertural, C.; Stoffel, R. P.; Müller, P. C.; Vogt, C. A.; Dronskowski, R., First-Principles Plane-Wave-Based Exploration of Cathode and Anode Materials for Li and Na-ion Batteries involving Complex Nitrogen-based Anions. *Chem. Mater.* **2022**, 34, 652–668.

COHP: Deintercalation of Li/Na dicyanamide salts

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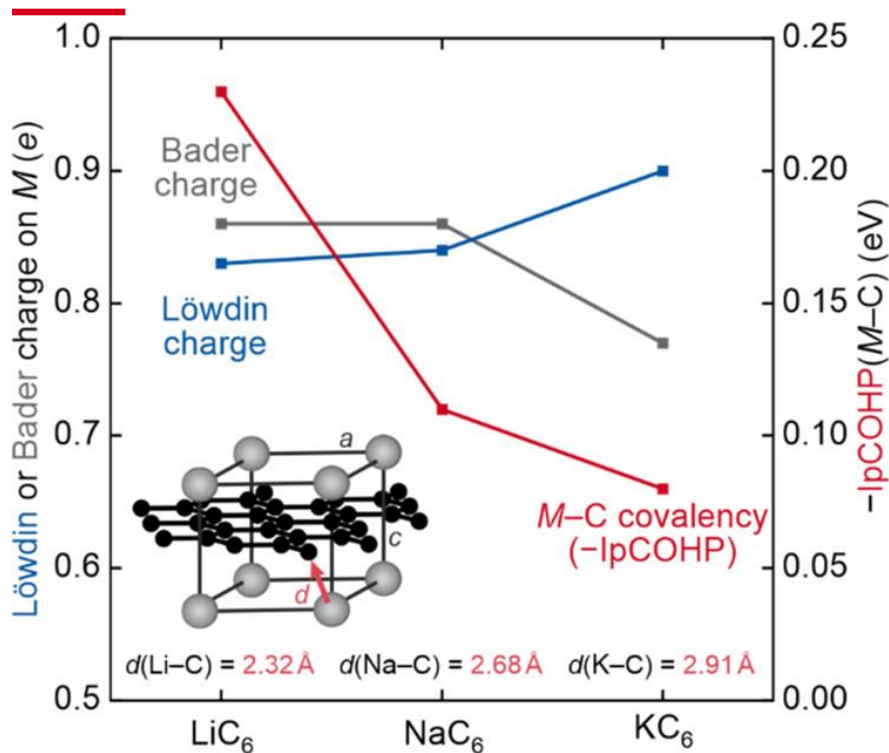
almost no
change >



Ertural, C.; Stoffel, R. P.; Müller, P. C.; Vogt, C. A.; Dronskowski, R., First-Principles Plane-Wave-Based Exploration of Cathode and Anode Materials for Li and Na-ion Batteries involving Complex Nitrogen-based Anions. *Chem. Mater.* **2022**, 34, 652–668.

Charges combined with COHP

- The intercalation problem of Na in (pristine) graphite



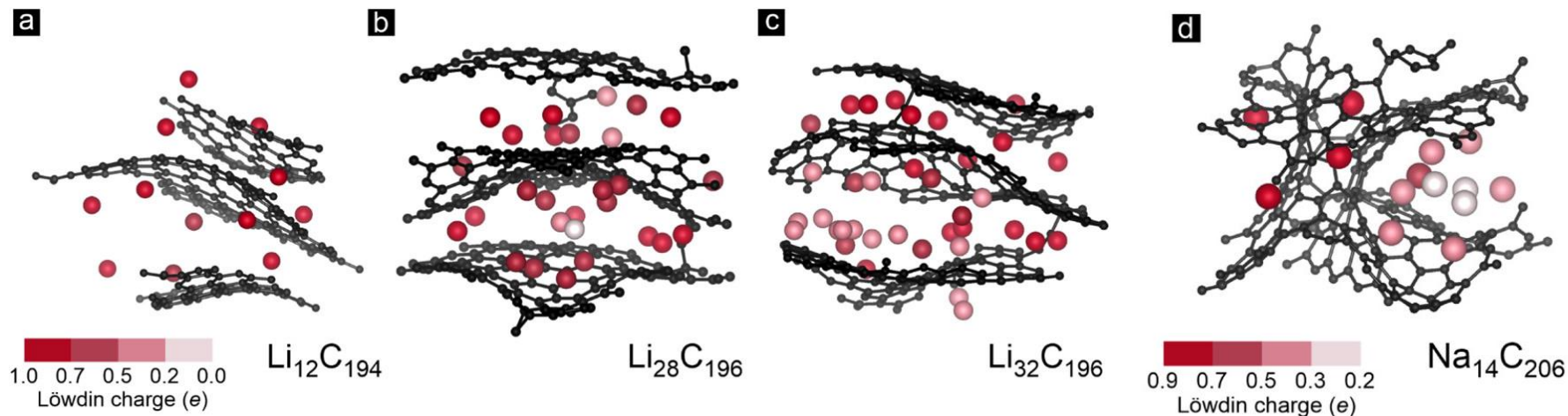
- > only Li is known to intercalate into pristine graphite, yielding LiC₆
- > Na does not intercalate at all
- > K favors the composition KC₈ (but stability issues known)
- > **why?**

Literature: LiC₆ is stabilized by an additional covalent contribution in M-C binding energy

- > confirmed by present study

Mulliken & Löwdin charges

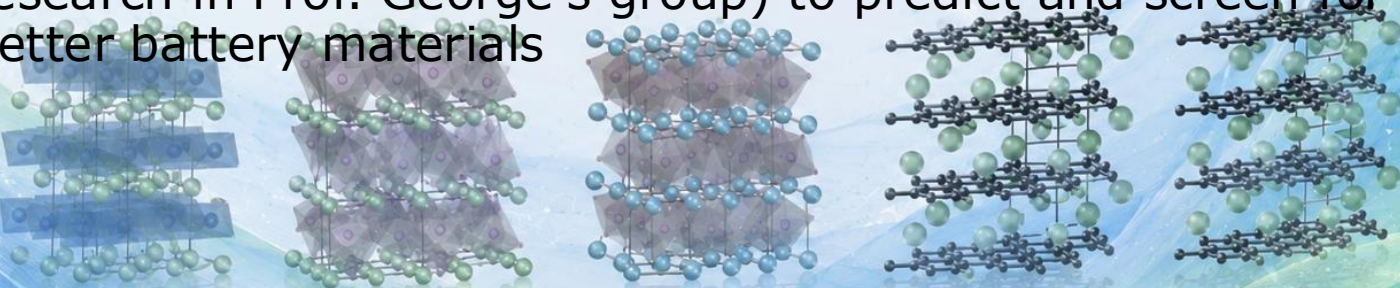
- Amorphous carbonaceous materials as alternative to Li/graphite anodes and to solve the intercalation problem of Na in (pristine) graphite



> good Coulombic (charging/discharging) behaviour
expected estimated from the Löwdin charges

Summary

- Commercial battery materials were examined using chemical bonding indicators
- Insights were used to look for alternative materials
 - transition metal free cathode materials
 - graphite-like amorphous anode materials to intercalate Na
- **Future steps:** Use automation workflows and quantum chemical bonding descriptors to train machine learning models (ongoing research in Prof. George's group) to predict and screen for new and better battery materials



Summary

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- **Future steps:** Use automation workflows and quantum chemical bonding descriptors to train machine learning models (ongoing research in Prof. George's group) to predict and screen for new and better battery materials

Thank you for your kind attention!

Check out quantumchemist.de (or scan the QR code) to learn more about my background!

